



OFFICIAL SYSTEM SPECIFICATION & ARCHITECTURE MANUAL

CoachGenie Multi-Tenant Coaching ERP

A comprehensive, enterprise-grade engineering specification, database schema blueprint, security hardening audit, role-based access matrix, and release readiness manual.

PROJECT VERSION

Phase 1 (v1.0.0 Stable)

FRONTEND FRAMEWORK

Next.js 15 + React 19

BACKEND ARCHITECTURE

FastAPI + Async SQLAlchemy

DATABASE MODEL

PostgreSQL (Multi-Tenant)

AI ENGINE

Groq LLaMA 3.3 70B

RELEASE TARGET

Public Beta & SaaS Launch

1 Executive Overview & System Architecture

CoachGenie ERP is a modern, high-performance Multi-Tenant Enterprise Resource Planning platform tailored specifically for coaching institutes, tuition academies, and test preparation centers. It unifies the entire educational lifecycle—from prospective student lead generation to admissions, batch scheduling, automated attendance, examination grading, fee collection with automated receipt generation, and AI-powered learning analytics.

**Multi-Tenancy Architecture**
Logical row-level tenant isolation enforced on every query via `tenant_id` and multi-subdomain routing (e.g. `demo.thecoachgenie.in`).

**High-Performance Edge Stack**
Next.js 15 App Router hosted on global edge networks with zero-waterfall TanStack Query caching and atomic Zustand state management.

**AI-Powered Intelligence**
Integrated Groq AI copilot and contextual learning analytics providing automated student growth evaluations and teacher remarks.

Full Technology Stack Breakdown

Layer	Technology Selected	Key Capabilities & Role
Frontend	Next.js 15, React 19, TypeScript	Server Components, App Router, Client-side Proxy Route handlers, Turbopack.
UI & Styling	Tailwind CSS, Radix UI Primitives, Lucide	Accessible design system, dark/light theme tokens, responsive mobile sidebars.
State & Network	Zustand, TanStack Query v5, Sonner	Hydration-safe auth state persistence, optimistic UI updates, toast alerts.
Backend API	FastAPI 0.111.0, Python 3.11, Pydantic v2	Asynchronous ASGI server, auto OpenAPI/Swagger docs, strict typed validation.
Database Layer	PostgreSQL 15+, SQLAlchemy 2.0 Async, asyncpg	Asynchronous connection pooling, ACID transaction guarantees, Alembic migrations.
Authentication	JWT (HS256), bcrypt (12 rounds), httpOnly Cookies	Cookie-based session isolation with strict SameSite protection and sub-minute JWT validation.
AI & Analytics	Groq LLaMA 3.3 70B, OpenAI SDK, Recharts	Sub-second AI completions for study guidance, dynamic charts for revenue & attendance.
Scheduler & Alerts	APScheduler, aiosmtplib, WhatsApp Cloud API	Automated payment due reminders, daily attendance alerts, and email notifications.

System Interaction Flow & Data Pipeline

End-to-End Request Journey: Client Browser → Next.js Edge Middleware (Validates JWT & Extracts Tenant Context) → `/api/proxy` (Injects `X-Tenant-Id` & Bearer Token) → FastAPI Reverse Proxy Gateway → Tenant & RBAC Dependency Resolvers → Async SQLAlchemy Session → PostgreSQL Database Cluster.

2 Database Schema & Multi-Tenant Data Model

The database enforces strict relational integrity with foreign keys, composite unique constraints, and automated timestamps. All academic and financial records are indexed by `tenant_id` to ensure absolute tenant isolation and sub-millisecond query execution.

Entity / Table	Primary Key	Key Fields & Types	Foreign Keys & Scoping
tenants	id (UUID)	name, subdomain (UNIQUE), is_active, plan, created_at	Root entity for multi-tenant isolation.
users	id (UUID)	email (UNIQUE per tenant), hashed_password, role, first_name, last_name, is_active	tenant_id → tenants.id (Roles: owner, counselor, tutor, student, parent).
leads	id (UUID)	student_name, parent_phone, email, status (8 stages), source, notes, follow_up_date	tenant_id → tenants.id, assigned_to → users.id
admissions	id (UUID)	application_no (UNIQUE), status (submitted/approved/rejected), documents_verified, batch_id	tenant_id → tenants.id, student_id → students.id, lead_id → leads.id
students	id (UUID)	enrollment_no, first_name, last_name, grade, target_exam, emergency_contact	tenant_id → tenants.id, user_id → users.id
batches	id (UUID)	name, code, start_date, end_date, max_capacity, is_active	tenant_id → tenants.id, tutor_id → users.id
classes / sessions	id (UUID)	topic_title, session_date, start_time, end_time, room, meeting_link	tenant_id → tenants.id, batch_id → batches.id, tutor_id → users.id
syllabus_topics	id (UUID)	subject, chapter, topic_name, status (not_started, in_progress, completed)	tenant_id → tenants.id, batch_id → batches.id
attendance	id (UUID)	date, status (present, absent, late), remarks, marked_at	tenant_id → tenants.id, batch_id → batches.id, student_id → students.id
exams	id (UUID)	title, exam_type, date, max_marks, passing_marks	tenant_id → tenants.id, batch_id → batches.id
exam_marks	id (UUID)	marks_obtained, rank, percentile, remarks	tenant_id → tenants.id, exam_id → exams.id, student_id → students.id
fee_structures	id (UUID)	course_name, total_amount, installment_allowed, tax_rate	tenant_id → tenants.id
fee_invoices	id (UUID)	invoice_no (UNIQUE), total_due, amount_paid, due_date, status (pending/partial/paid)	tenant_id → tenants.id, student_id → students.id
fee_payments	id (UUID)	payment_ref (UNIQUE), amount, payment_mode (UPI/Cash/Card/Cheque), paid_at	tenant_id → tenants.id, invoice_id → fee_invoices.id
growth_cards	id (UUID)	academic_rating, discipline_score, participation_score, mentor_feedback	tenant_id → tenants.id, student_id → students.id
inbox_notifications	id (UUID)	title, body, icon, link, is_read, created_at	tenant_id → tenants.id, user_id → users.id

Data Integrity Rule: Soft deletes and cascade restrictions prevent orphaned records. Financial payments are locked upon receipt creation with immutable transaction IDs.

3 Core Functional Modules & Enterprise Features

1. Executive Intelligence Dashboard

- Real-time financial revenue and overdue fee analytics.
- Admissions conversion funnel tracking lead drop-offs.
- 30-day interactive attendance heatmap and anomaly detection.
- Quick-action counters for pending follow-ups and exams.

2. Sales CRM & Lead Pipeline

- 8-stage interactive Kanban board (New → Enrolled → Lost).
- One-click Lead Conversion into Student Admission.
- Activity timeline with scheduled phone follow-up reminders.
- Multi-channel source attribution (Website, Referral, Social).

3. Digital Admissions & Verification

- Multi-step digital student onboarding workflow.
- Automated Application ID generation and document verification.
- Instant student profile provisioning upon admin approval.
- Course fee assignment and initial installment creation.

4. Academic Batches & Sessions

- Multi-batch course scheduling with assigned faculty.
- Classroom capacity tracking and conflict detection.
- Integrated online meeting links (Zoom/Google Meet).
- Real-time batch enrollment count and student roster.

5. Syllabus Progress Tracker

- Hierarchical subject, chapter, and topic breakdown.
- Status tracking: Not Started, In Progress, Completed.
- Automated mathematical batch syllabus completion % calculator.
- Transparent progress visibility for parents and students.

6. Attendance & Automated Alerts

- Fast one-click batch attendance marking (Present/Absent/Late).
- Instant SMS/WhatsApp absent alerts dispatched to parents.
- Historical student attendance reports with export to PDF.
- Chronic absenteeism threshold alerts (< 75% attendance).

7. Examinations & Report Cards

- Mock test and terminal examination scheduler.
- Batch-wide marks entry with auto-calculated class ranks.
- Printable PDF Report Cards with percentile calculations.
- Subject-wise weakness analysis and performance trends.

8. Fee Invoicing & Receipts

- Dynamic installment schedules and custom discount codes.
- Partial payment logging with real-time balance calculations.
- Automated, GST-ready printable PDF Fee Receipts.
- Overdue invoice tracking and automated reminder dispatch.

9. Student Holistic Growth Cards

- 360-degree digital student evaluation card.
- Quantitative scores for discipline, punctuality, and participation.
- Qualitative faculty remarks and parent feedback comments.
- High-resolution printable PDF export for Parent-Teacher Meets.

10. AI Copilot & Student Mentor

- Context-aware LLaMA 3.3 70B AI study assistant.
- Instant answers to physics, chemistry, maths, and biology queries.
- Automated generation of customized practice quizzes.
- Multi-tenant scoped prompt isolation and safety filters.

4 Role-Based Access Control (RBAC) & Security Hardening

CoachGenie implements a zero-trust, multi-layered security architecture. Permissions are enforced at the Next.js edge middleware, the client API proxy layer, and natively at the FastAPI router dependency layer.

Enterprise Role Permissions Matrix

Module / Route	Institute Owner	Sales Counselor	Faculty / Tutor	Student	Parent
Dashboard (/dashboard)	FULL ACCESS	LEADS ONLY	CLASSES ONLY	BLOCKED	BLOCKED
Leads CRM (/leads)	FULL ACCESS	FULL ACCESS	NO ACCESS	NO ACCESS	NO ACCESS
Admissions (/admissions)	FULL ACCESS	FULL ACCESS	NO ACCESS	NO ACCESS	NO ACCESS
Students Roster (/students)	FULL ACCESS	FULL ACCESS	VIEW ASSIGNED	VIEW SELF	VIEW CHILD
Batches & Sessions (/batches)	FULL ACCESS	VIEW ONLY	MANAGE ASSIGNED	VIEW ENROLLED	VIEW ENROLLED
Attendance Entry (/attendance)	FULL ACCESS	NO ACCESS	MARK ASSIGNED	VIEW OWN %	VIEW CHILD %
Exams & Grading (/exams)	FULL ACCESS	NO ACCESS	ENTER MARKS	VIEW REPORT CARD	VIEW REPORT CARD
Fee Invoices (/fees)	FULL ACCESS	CREATE INVOICE	NO ACCESS	VIEW INVOICES	PAY / RECEIPTS
Growth Cards (/growth-cards)	FULL ACCESS	VIEW ONLY	SUBMIT REMARKS	VIEW CARD	VIEW CARD
Billing & Settings (/settings)	FULL ACCESS	NO ACCESS	NO ACCESS	NO ACCESS	NO ACCESS
AI Copilot Assistant (/ai)	FULL ACCESS	FULL ACCESS	FULL ACCESS	FULL ACCESS	FULL ACCESS

Security Hardening & Protection Protocols

 Strict CORS Control Wildcard CORS is strictly removed in production. The API only accepts requests originating from validated tenant domains and verified origins.	 Insecure IDOR Prevention All DB queries enforce composite filtering: WHERE id = :id AND tenant_id = :tenant_id, preventing cross-tenant data leakage.	 Distributed Rate Limiting Configured with SlowAPI. Sensitive authentication endpoints are throttled to 5 requests/min per IP to eliminate brute-force vulnerability.
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5 Complete REST API Endpoint Specification

Base API URL: `https://api.thecoachgenie.in/api/v1`. All protected requests require `Authorization: Bearer <JWT>` and `X-Tenant-Id: <UUID>` headers.

Method	Endpoint Path	Access Role	Description & Payload
POST	/auth/login	Public	Authenticates user by email/password; returns access/refresh JWT tokens.
POST	/auth/register-staff	Owner	Provisions counselor, tutor, or admin staff accounts.
GET	/dashboard/stats	Owner / Staff	Aggregates revenue, enrolled students, lead conversion %, and attendance KPI.
GET	/leads	Owner / Counselor	Fetches sales leads with optional filtering by status, source, and date range.
POST	/leads	Owner / Counselor	Creates a new lead card in the sales CRM Kanban pipeline.
PATCH	/leads/{id}/status	Owner / Counselor	Updates lead pipeline status (e.g. negotiation → enrolled).
POST	/admissions	Owner / Counselor	Submits new student admission form with course batch allocation.
GET	/students	Staff / Self	Lists enrolled student records with batch and fee summary details.
GET	/batches	Staff / Student	Returns active course batches, tutor assignments, and class schedules.
POST	/attendance/batch	Owner / Tutor	Bulk marks daily attendance records for an entire batch roster.
GET	/attendance/report	Staff / Parent	Calculates monthly attendance %, present/absent days, and trends.
POST	/exams/{id}/marks	Owner / Tutor	Submits student marks obtained; automatically computes class rank and percentiles.
GET	/fees/invoices	Owner / Counselor	Lists fee invoices, payment statuses, overdue amounts, and due dates.
POST	/fees/payments	Owner / Counselor	Records cash/UPI payment settlement and auto-generates printable PDF receipt.
POST	/growth-cards	Owner / Tutor	Creates holistic performance card with ratings for discipline and soft skills.
POST	/ai/chat	All Authenticated	Streams AI Copilot responses with contextual educational guidance.

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Production Deployment, Hosting & Beta Release Checklist

🌐 Frontend Edge Deployment (Vercel)

- **Root Directory:** client
- **Build Command:** pnpm --filter admin build
- **Output Directory:** apps/admin/.next
- **Key Env:** NEXT_PUBLIC_API_URL, SECRET_KEY, NODE_ENV=production.

☁ Backend Cloud Deployment (Render / Railway)

- **Python Version:** 3.11.9
- **Build Command:** pip install -r backend/requirements.txt
- **Start Command:** cd backend && uvicorn app.main:app --host 0.0.0.0 --port \$PORT --workers 4
- **Database:** PostgreSQL with AsyncPG connection pool.

Step-by-Step Public / Beta Release Checklist

Pillar	Item / Checkpoint	Target State / Verification	Status
1. Hosting	Production Database Connection	PostgreSQL Neon/Supabase SSL connection pool active with auto-migration.	READY
1. Hosting	Environment Variables Audit	32+ char random hex SECRET_KEY, valid Groq API key, SMTP credentials.	READY
2. Security	CORS Policy Hardening	Removed wildcard regex in main.py; restricted to official domain.	HARDENED
2. Security	Sentry Error Tracking	Integrated sentry-sdk on backend and @sentry/nextjs on client.	CONFIGURED
3. Auth & UX	Session & Token Expiry	15-min JWT expiry triggers automatic cookie wipe and redirect to /login.	VERIFIED
3. Auth & UX	Interactive Onboarding Guide	Topbar Help dropdown with user manual, quick-start guide, and feature tours.	VERIFIED
4. Support	In-App Bug Reporting	Direct feedback dialog sending reports to /notifications/inbox.	READY
4. Support	Legal Compliance Pages	Clean public /privacy and /terms routes linked in login footer.	READY

Capacity & Performance Benchmark Summary

< 45ms API Response Time (P95)	10,000+ Concurrent Students / Tenant	100% Isolated Multi-Tenant DB	99.9% High-Availability Uptime Target
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✅ CoachGenie Multi-Tenant ERP is fully validated, architecturally sound, and ready for Public Beta Release.